PRODUCT SPECIFICATION

EPD NO: MF-17506-M009F-T

CUSTOMER: _____

APPROVED BY CUSTOMER	
Approved by	Remark

APPROVED BY Ink Cubic			
Prepared by	Checked by		Approved by
	EE	ME	
Yuchen Wang	Chuanqing Wu	Hao Tong	Zheng Han

Version	Content	Date	Producer
A0	New release	2025/12/30	Wang
A1	Modify Mechanical Drawing of EPD module	2026/2/28	Tong

CONTENTS

1. General Description 4
2. Features 4
3. Application 4
4. Mechanical Specifications 4
5. Mechanical Drawing of EPD module 5
6. Input/Output Terminals 6
7. MCU Interface 8
   7.1 MCU interface selection 8
   7.2 MCU Serial Peripheral Interface (4-wire SPI) 8
   7.3 MCU Serial Peripheral Interface 3-wire SPI) 9
8. Reference Circuit 10
9. Absolute Maximum Rating 11
10. DC Characteristics 11
11. Serial Peripheral Interface Timing and Command Table 13
   11.1 Serial Peripheral Interface Timing 13
   11.2 Command Table 14
12. Optical characteristics 15
   12.1 Specifications 15
   12.2 Definition of contrast ratio 16
   12.3 Reflection Ratio 16
13. Handling, Safety and Enviromental Requirements 17
14. Reliability test 19
15. Block Diagram 20
16. PartA/PartB Specification 20
17. Point and line standard 21
18. Barcode 23
19. Packing 24

1. General Description

Ink Cubic-M009F-T is an Active Matrix Electrophoretic Display (AMEPD), with interface and a reference system design. The 7.50'' active area contains 1024×600 pixels, and has 3-bit E6 full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC, SRAM, LUT, VCOM and border are supplied with each panel.

2. Features

- 1024× 600 pixels display
- High contrast
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Low voltage detect for supply voltage
- High voltage ready detect for driving voltage
- Internal temperature sensor
- Waveform stored in On-chip OTP
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage

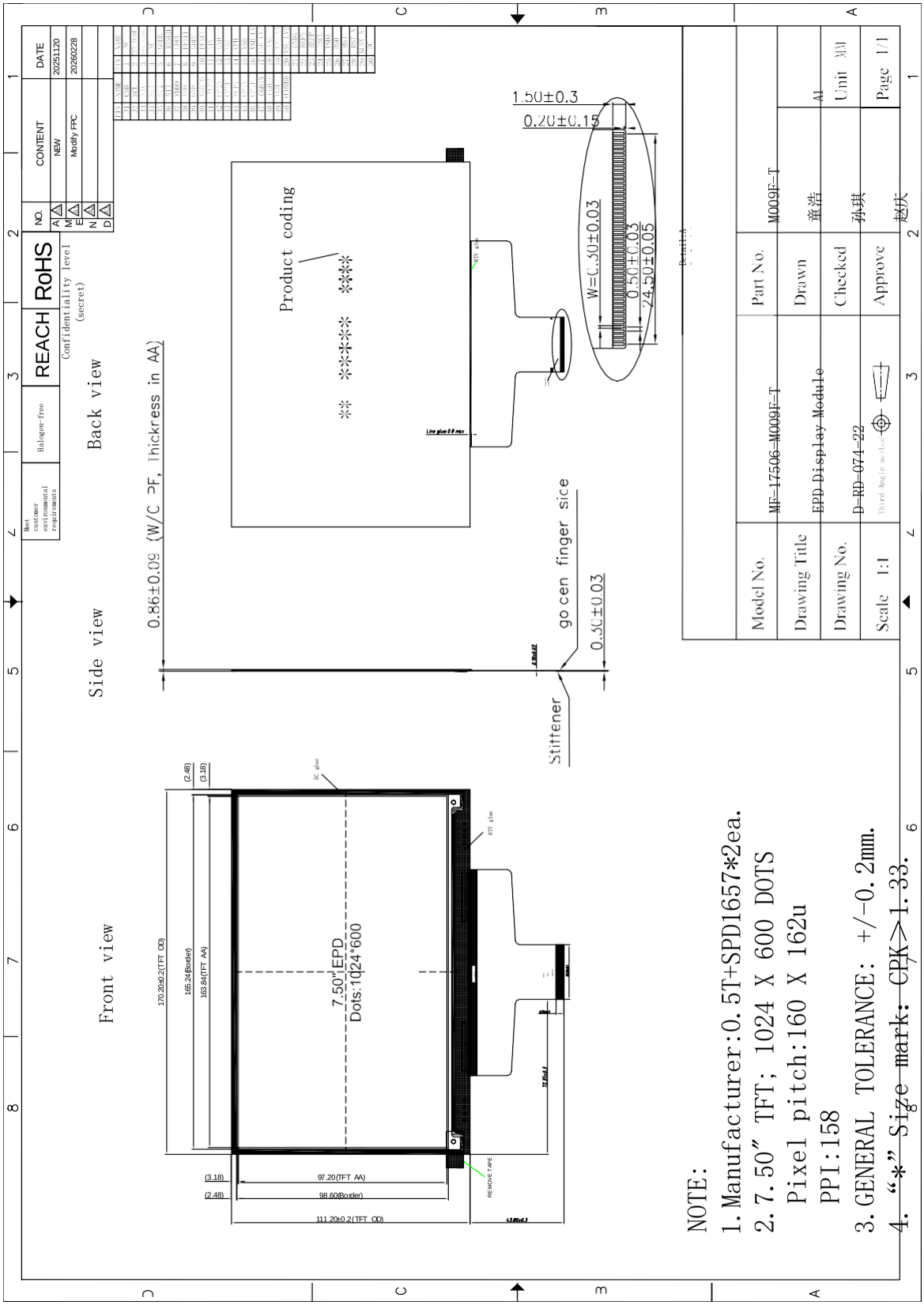
3. Application

Electronic desk nameplate etc.

4. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	7.50	Inch	
Display Resolution	1024(H)×600(V)	Pixel	Dpi:158
Active Area	163.84(H)×97.20(V)	mm	
Pixel Pitch	0.160×0.162	mm	
Pixel Configuration	Rectangle		
Outline Dimension	170.20(H)×111.20 (V) ×0.86(D)	mm	
Weight	TBD	g	

5.Mechanical Drawing of EPD module



6. Input/Output Terminals

Pin #	Single	Description	Remark
1	NC	No connection and do not connect with other NC pins	
2	TFT_VCOM	TFT_VCOM driving voltage	
3	FPL_VCOM	FPL_VCOM driving voltage	
4	SL_CSB	Slave Chip Select input pin (SL_CSB)	
5	GDRH	N-Channel MOSFET Gate Drive Control	
6	RESEH	Current Sense Input for the Control Loop	
7	GDRL	Reserved	
8	RESEL	Ground	
9	GDRC	P-Channel MOSFET Gate Drive Control	
10	RESEC	Current Sense Input for the Control Loop	
11	VPC	VPC driving voltage	
12	GND	Ground	
13	VGL	Negative Gate driving voltage	
14	VPH	VPH driving voltage	
15	VSH	Positive Source driving voltage	
16	VSH_LV	Positive Source driving voltage	
17	VSH_LV2	Positive Source driving voltage	
18	VSL	Negative Source driving voltage	
19	VSL_LV	Negative Source driving voltage	
20	VSL_LV2	Negative Source driving voltage	
21	GNDA	Ground ; Connect to GND	
22	REFN	Reserved	
23	REFP	Reserved	
24	TSCL	Cascade signal, connect a pull-up resistor	
25	TSDA	Cascade signal, connect a pull-up resistor	
26	BS0	Bus selection pin; L: 4-wire IF. H: 3-wire IF.	
27	BS1	Bus selection pin; L: refer to BS0. H: Standard 4-wire SPI/dual SPI/quad SPI	
28	RST_N	Reset	
29	BUSY_N	Busy state output pin	
30	DC	Data /Command control pin (D/C)	

31	CSB	Chip Select input pin (CSB)	
32	SCL	Serial clock pin (SPI)	
33	SI0	serial data pin (SPI)	
34	SI1	serial data pin ; Reserved	
35	SI2	serial data pin ; Reserved	
36	SI3	serial data pin ; Reserved	
37	VDDD	Core logic power pin; Connect to VDDD	
38	VDD	Supply voltage	
39	GND	Ground; Connect to GNDA	
40	VDDIO	Supply voltage	
41	VCP2	Charge Pump Pin	
42	CP2N	Charge Pump Pin	
43	CP2P	Charge Pump Pin	
44	VCP1	Charge Pump Pin	
45	CP1N	Charge Pump Pin	
46	CP1P	Charge Pump Pin	
47	CGH1N	Charge Pump Pin; Reserved	
48	CGH1P	Charge Pump Pin; Reserved	
49	VGH	Positive Gate driving voltage	
50	VCOMBD	VCOMBD driving voltage	

Note 6-1: This pin (CSB) is the chip select input connecting to the MCU. The chip is enabled forMCU communication: only when CSB is pulled Low.

Note 6-2: This pin (DC) is Data/Command control pin connecting to the MCU. When the pin is pulled High, the data will be interpreted as data. When the pin is pulled Low, the data will be interpreted as command.

Note 6-3: This pin (RST_N) is reset signal input. The Reset is active low.

Note 6-4: This pin (BUSY_N) is Busy state output pin. When Busy is High ,the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put Busy pin High when the driver IC is working such as:

- Outputting display waveform; or
- Communicating with digital temperature sensor

Note 6-5: This pin (BS0) is for 3-line SPI or 4-line SPI selection. When it is “Low”, 4-line SPI isselected. When it is “High”, 3-line SPI (9 bits SPI) is selected.

7.MCU Interface

7.1 MCU interface selection

The Ink Cubic-M009F-T can support 3-wire/4-wire serial peripheral interface. In the Module, the MCU interface is pin selectable by BS1 pins shown in.

Table 7-1: Interface pins assignment under different MCU interface

	Pin Name						
MCU Interface	BS1	BS0	RST_N	CSB	DC	SCL	SI[0:3]
4-wire serial peripheral interface (SPI)	L	L	Required	CSB	D/C	SCL	SI0=SDA
3-wire serial peripheral interface (SPI) – 9 bits SPI	L	H	Required	CSB	L	SCL	SI0=SDA
Standard 4-wire SPI	H	L/H	Required	CSB	L	SCL	SI0: Serial SPI MOSI; Dual SPI MOSI; Quad SPI MOSI SI1: Serial SPI MISO; Dual SPI SI1; Quad SPI SI1 SI2: Serial SPI L; Dual SPI L; Quad SPI SI2 SI3: Serial SPI L; Dual SPI L; Quad SPI SI3

7.2 MCU Serial Peripheral Interface (4-wire SPI)

The 4-wire SPI consists of serial clock SCL, serial data SDA, DC and CSB. The control pins status in 4-wire SPI in reading/writing command/data is shown in Table 7-2 and the read/write procedure 4-wire SPI is shown in Figue 7-2.

Table 7-2 : Control pins status of 4-wire SPI

Function	SCL pin	SDA pin	DC pin	CSB pin
Write command	↑	Command bit	L	L
Read/Write data	↑	Data bit	H	L

Note:

(1) L is connected to VSS and H is connected to VDDIO.

(2) ↑ stands for rising edge of signal.

(3) SDA (Write Mode) is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/Cshould be kept over the whole byte. The data byte in the shift register is written to the Graphic Display Data RAM (RAM)/Data Byte register or command Byte register according to DC pin.

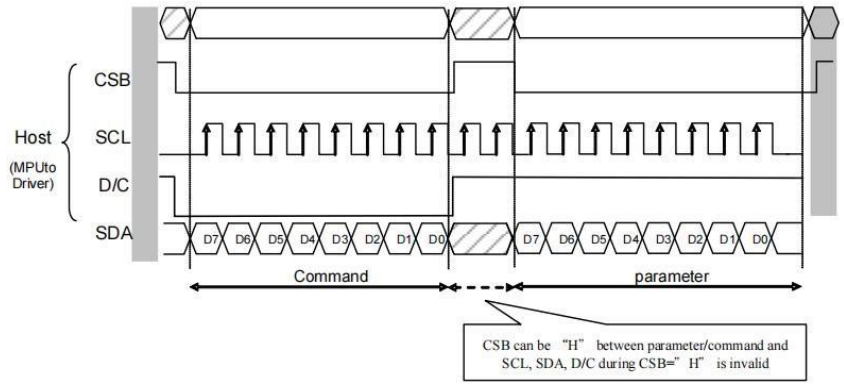


Figure 7-1: Read/Write procedure in 4-wire SPI mode

7.3 MCU Serial Peripheral Interface (3-wire SPI)

The 3-wire SPI consists of serial clock SCL, serial data SDA and CSB. The operation is similar to 4-wire SPI while DC pin is not used and it must be tied to LOW. The control pins status in 3-wire SPI is shown in Table 7-3.

Table 7-3 : Control pins status of 3-wire SPI

Function	SCL pin	SDA pin	DC pin	CSB pin
Write command	↑	Command bit	Tie LOW	L
Write data	↑	Data bit	Tie LOW	L

Note:

(1) L is connected to V_{SS} and H is connected to V_{DDIO}

(2) ↑ stands for rising edge of signal

In the read/write operation, a 9-bit data will be shifted into the shift register on every clock rising edge. The bit shifting sequence is DC bit, D7 bit, D6 bit to D0 bit. The first bit is DC bit which determines the following byte is command or data. When DC bit is 0, the following byte is command. When DC bit is 1, the following byte is data. The read/write procedure of 3-wire SPI is shown in Figure 7-2.

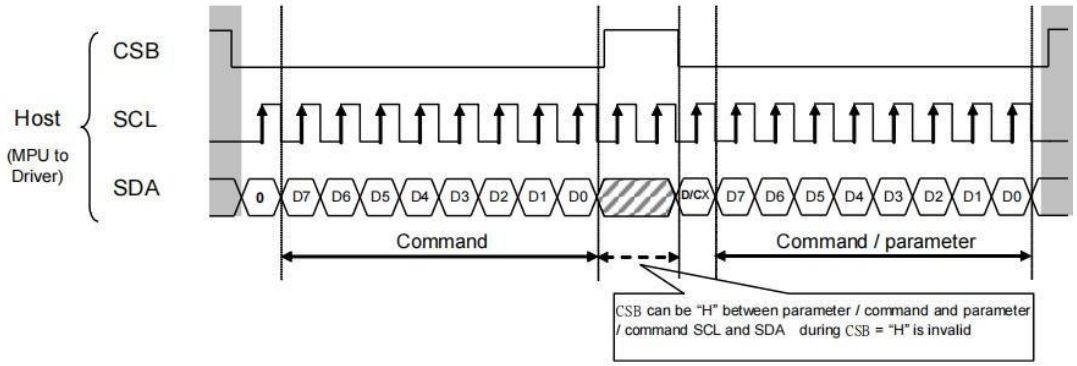
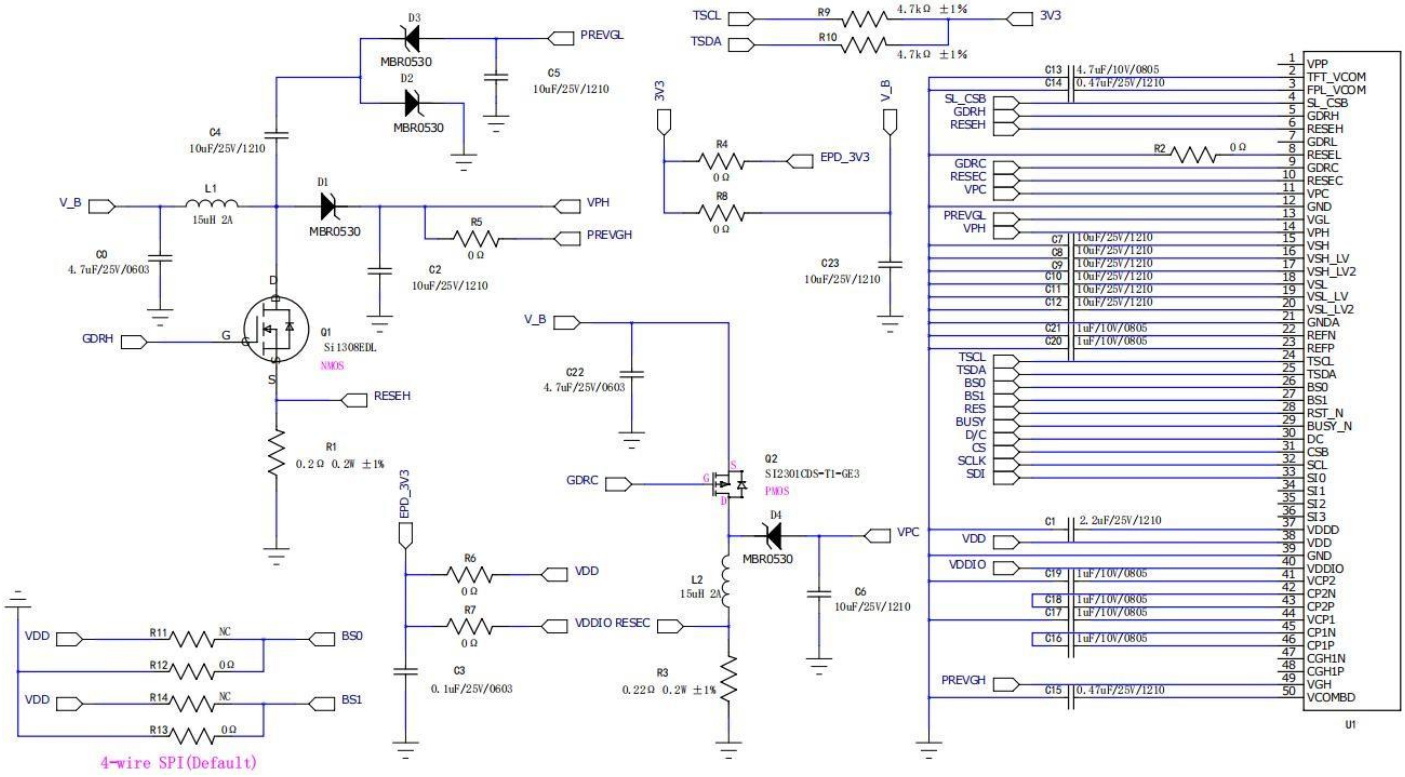


Figure 7-2: Write procedure in 4-wire SPI mode

8.Reference Circuit



Part Name	SPD1657B Value /quirement/Reference Part
C0,C22	4.7uF/0603/X5R;Voltage Rating: 25V
C1	2.2uF/0603/X7R;Voltage Rating: 25V
C2,C4-C12,C23	10uF/1210/X7R;Voltage Rating: 25V
C3	0.1uF/0603/X7R;Voltage Rating: 25V
C13	4.7uF/0805/X7R;Voltage Rating: 10V
C14-C15	0.47uF/1210/X7R;Voltage Rating: 25V
C16-C21	1uF/0805/X7R;Voltage Rating: 10V
L1-L2	15uH;2A
Q1	NMOS VDS=30V， Ron≤77mΩ@2.5V VGS，Typ Qg≤3.5 nC
Q2	PMOS VDS=-20V，Ron≤85mΩ@-2.5V VGS，Typ Qg≤7.6 nC
D1-D4	MBR0530
R1	0.2Ω;0.2W
R2,R4-R8,R12-R13	0Ω
R3	0.22Ω;0.2W
R9-R10	4.7kΩ;0.1W,±1%

9. Absolute Maximum Rating

Table 9-1: Maximum Ratings

Symbol	Parameter	Rating	Unit
V_{CI}	Logic supply voltage	-0.5 to +3.6	V
T_{OPR}	Operation Temperature range	0 to 50	℃
Tst	Storage Temperature range	-25 to 60	℃
$Tstg_o$	Optimal Storage Temperature	23 ± 2	℃
$Hstg_o$	Optimal Storage Humidity	55 ±10	%RH

Note: Absolute Maximum Ratings are stress ratings. Stresses in excess of these ratings can cause permanent damage to the device. Functional operation of the device at these orany other conditions beyond those indicated in the operational sections of this data sheet is not implied. Exposing device to the absolute maximum ratings in a long period of timemay degrade the device and affect its reliability.

10. DC Characteristics

The following specifications apply for: VSS=0V, VDD=3.0V, T_{OPR}=25°C.

Table 10-1: DC Characteristics

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
V_{DD}	V_{DD} operation voltage	-	2.4	3.0	3.6	V
V_{IH}	High level input voltage	-	$0.8V_{DDIO}$		V_{DDIO}	V
V_{IL}	Low level input voltage	-	GND	-	$0.3V_{DDIO}$	V
V_{OH}	High level output voltage	Iout= 1mA VDD=2.4V	$0.8V_{DDIO}$	-	-	V
V_{OL}	Low level output voltage	Iout= 1mA VDD=2.4V	GND	-	GND+0.4	V
Iupdate	Module operating current	-	-	TBD	-	mA
Ist_V_{DD}	Digital Stand-by Current (power off mode)	-	-	1	-	uA

Notes 10-1: The Module operating current data is measured by using Oscilloscope, and extract the Mean value.

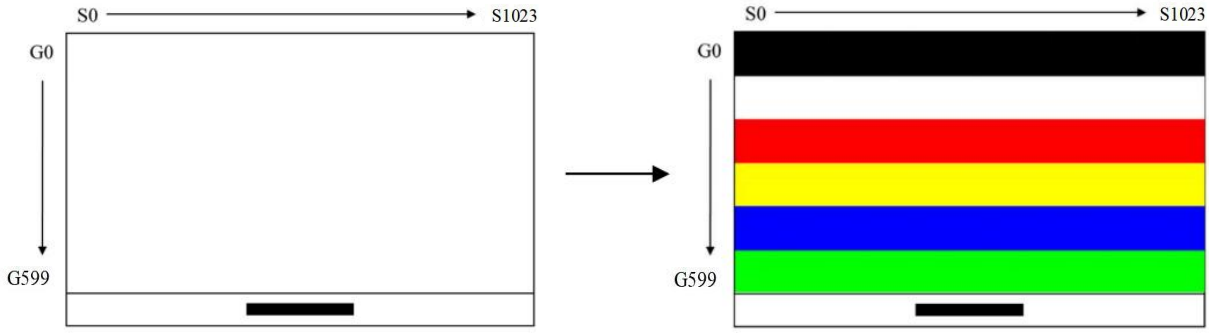
Notes 10-2: The typical power consumption is measured using associated 25C waveform with following pattern transition: from full white pattern to color stripe pattern.（PIC 10-1）

Notes 10-3: The high loading power consumption is measured using associated 25C waveform with following pattern transition: from full white pattern to noise pattern (including random scattering of 4 colors) .（PIC 10-2）

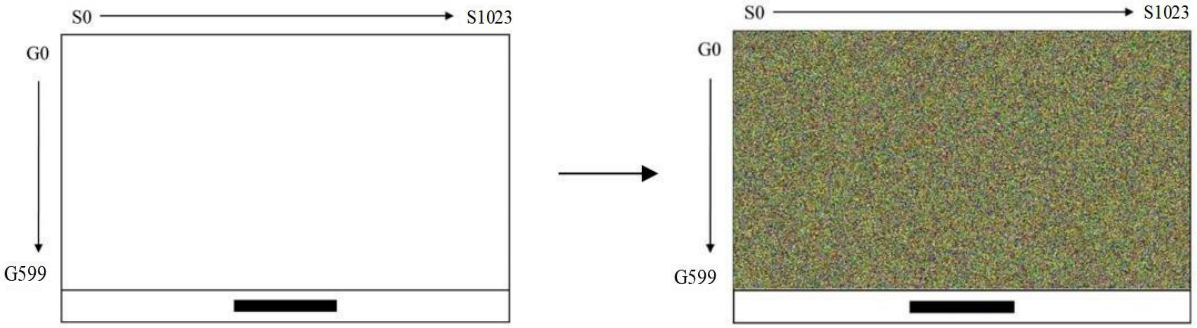
Notes 10-4: The minimum VDD value by 2.4V is based on typical application pattern with stable and continuing power supply. It does not apply on high loading pattern.（PIC 10-2）

Notes 10-5: The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by InkcCubic.

Notes 10-6: Vcom value has been set in the IC chip on the panel.



Picture 10-1:The typical power consumption



Picture 10-2:The high loading power consumption

11. Serial Peripheral Interface Timing and Command Table

11.1 Serial Peripheral Interface Timing

The following specifications apply for: VSS=0V, VCI=2.3V to 3.6V, T_{OPR}=25℃

Serial Peripheral Interface Timing Characteristics

Symbol	Parameter	Min	Typ	Max	Unit
t_{CSS}	CSB select setup time	60			ns
t_{CSH}	CSB select hold time	65			ns
t_{SCC}	CSB deselect setup time	20			ns
t_{CHW}	CSB deselect hold time	40			ns
t_{SCYCW}	Serial clock cycle (Write)	50			ns
t_{SHW}	SCL “H” pulse width (Write)	25			ns
t_{SLW}	SCL “L” pulse width (Write)	25			ns
t_{SCYCL}	Serial clock cycle (Read)	150			ns
t_{SHR}	SCL “H” pulse width (Read)	60			ns
t_{SLR}	SCL “L” pulse width (Read)	60			ns
t_{SDS}	Data setup time	30			ns
t_{SDH}	Data hold time	30			ns
t_{ACC}	Access time			75	ns
t_{OH}	Output disable time	10			ns

Note: All timings are based on 20% to 80% of VDDIO-VSS

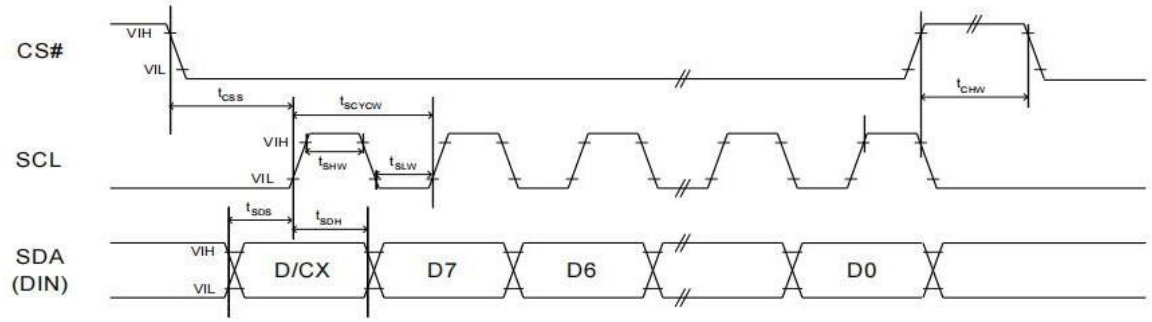


Figure 12-1: 3 pin serial interface characteristics (write mode)

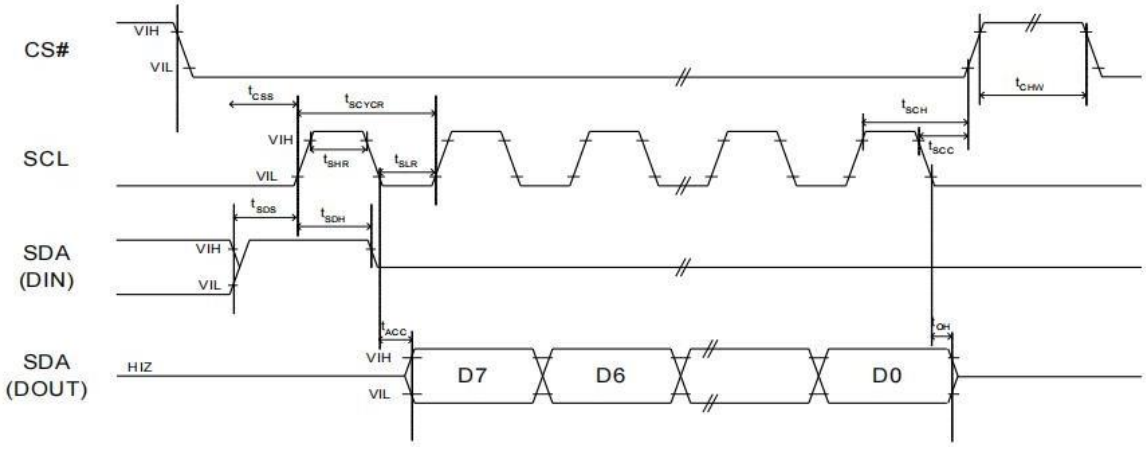


Figure 12-2: 3 pin serial interface characteristics (read mode)

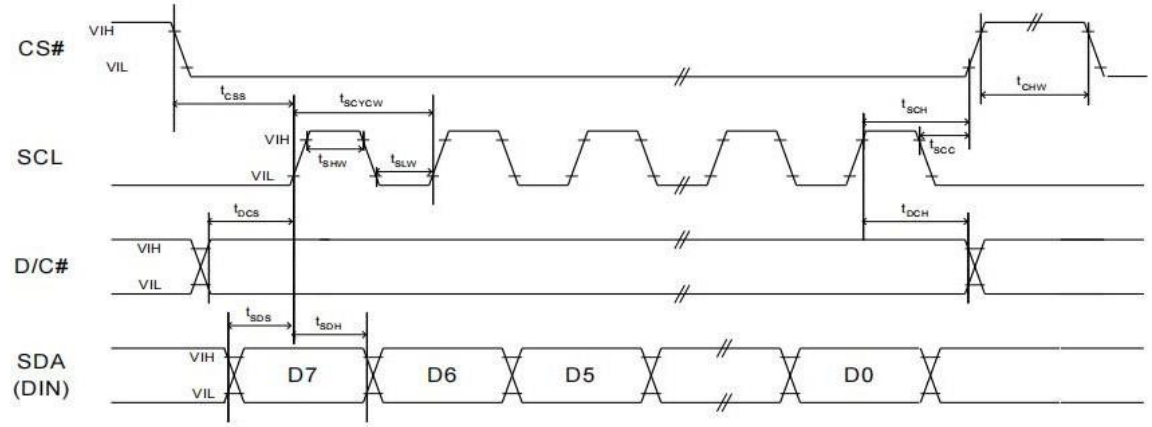


Figure 12-3: 4 pin serial interface characteristics (write mode)

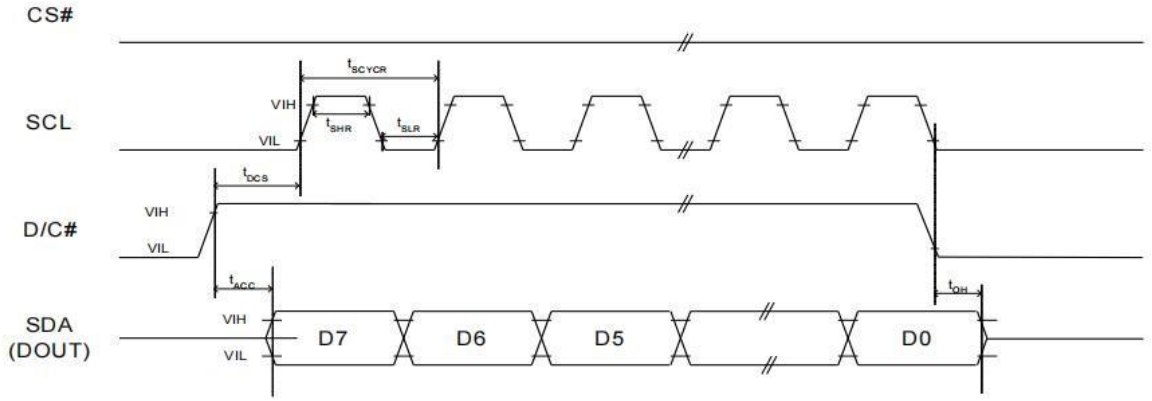


Figure 12-4: 4 pin serial interface characteristics (read mode)

11.2 Command Table

Please refer to IC SPD1657B datasheet.

12. Optical characteristics

12.1 Specifications

Measurements are made with that the illumination is under an angle of 45 degrees, the detection is perpendicular unless otherwise specified.

T=25℃

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP.	MAX	UNIT	Note
KS	Black State L* value	Black	-	-	16		Note 1
	Black State a* value	Black	-	-	15		Note 1
	Black State b* value	Black	-20	-	-		Note 1
WS	White State L* value	White	62	-	-		Note 1
	White State a* value	White	-7	-	1		Note 1
	White State b* value	White	-	-	3.5		Note 1
RS	Red State L* value	Red	23	-	-		Note 1
	Red State a* value	Red	36	-	-		Note 1
	Red State b* value	Red	-	-	38		Note 1
YS	Yellow State L* value	Yellow	56	-	-		Note 1
	Yellow State a* value	Yellow	-20	-	5		Note 1
	Yellow State b* value	Yellow	55	-	-		Note 1
BS	Blue State L* value	Blue	22	-	-		Note 1
	Blue State a* value	Blue	-	-	13		Note 1
	Blue State b* value	Blue	-	-	-29		Note 1
GS	Green State L* value	Green	30	-	-		Note 1
	Green State a* value	Green	-	-	-15		Note 1
	Green State b* value	Green	-5	-	30		Note 1

WS : White state, KS : Black state, RS: Red state， YS:Yellow state,BS： Blue state,GS:Green state;

Note 1 : Luminance meter : i - One Pro Spectrophotometer

Note 1:We don't guarantee 5 years pixels display quality for humidity below 45%RH or above 70%RH； Suggest Updated once a day;

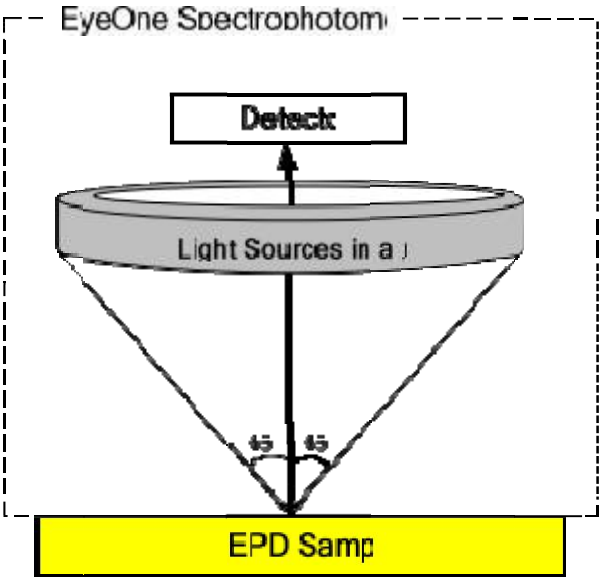
Note 1: To increases the black and white screen clear screen when red has refreshed for a long time , the effect is better.

12.2 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (R1) and the reflectance in a dark area (Rd)() :

R1: white reflectance Rd: dark reflectance

CR = R1/Rd

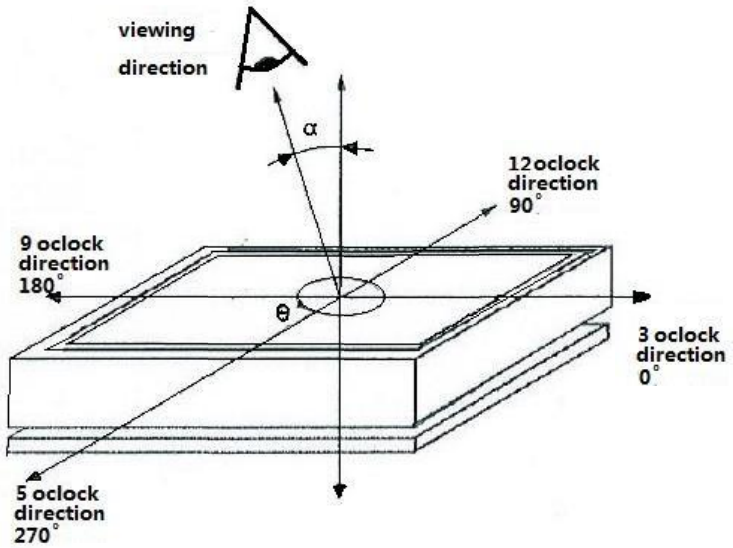


12.3 Reflection Ratio

The reflection ratio is expressed as:

R = Reflectance Factor $_{\text{white board}}$ x (L_{center} / $L_{\text{white board}}$)

L_{center} is the luminance measured at center in a white area (R=G =B=1). $L_{\text{white board}}$ is the luminance of a standard whiteboard. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.



13. Handling, Safety and Enviromental Requirements

WARNING
The display module should be kept flat or fixed to a rigid, curved support with limited bending along the long axis. It should not be used for continual flexing and bending. Handle with care. Should the display break do not touch any material that leaks out. In case of contact with the leaked material then wash with water and soap.

CAUTION
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.
Disassembling the display module can cause permanent damage and invalidate the warranty agreements.
IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.
Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged . Moreover the display is sensitive to static electricity and other rough environmental conditions.

Mounting Precautions
(1) It`s recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.
(2) It`s recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.
(3) You should adopt radiation structure to satisfy the temperature specification.
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.

Data sheet status	
Product specification	The data sheet contains final product specifications.

Limiting values

Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.

Application information

Where application information is given, it is advisory and dose not form part of the specification.

Product Environmental certification

ROHS

REMARK

All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.

14. Reliability test

	TEST	CONDITION	REMARK
1	High-Temperature Operation	T=40℃, RH=35%RH, For 240Hr	
2	Low-Temperature Operation	T = 0℃ for 240 Hr	
3	High-Temperature Storage	T=60℃ RH=35%RH For 240Hr	Test in white pattern
4	Low-Temperature Storage	T = -25℃ for 240 Hr	Test in white pattern
5	High Temperature, High-Humidity Operation	T=40℃,RH=80%RH, For 240Hr	
6	High Temperature, High-Humidity Storage	T=50℃,RH=80%RH,For 240Hr	Test in white pattern
7	Temperature Cycle	-25℃(30min)~60℃(30min),50 Cycle	Test in white pattern
8	Electrostatic discharge	HBM:330Ω,150pF	Air +/-4KV;Contact +/-2KV
9	UV Exposure Resistance	765 W/㎡ for 168hrs,40 ℃	

Actual EMC level to be measured on customer application.

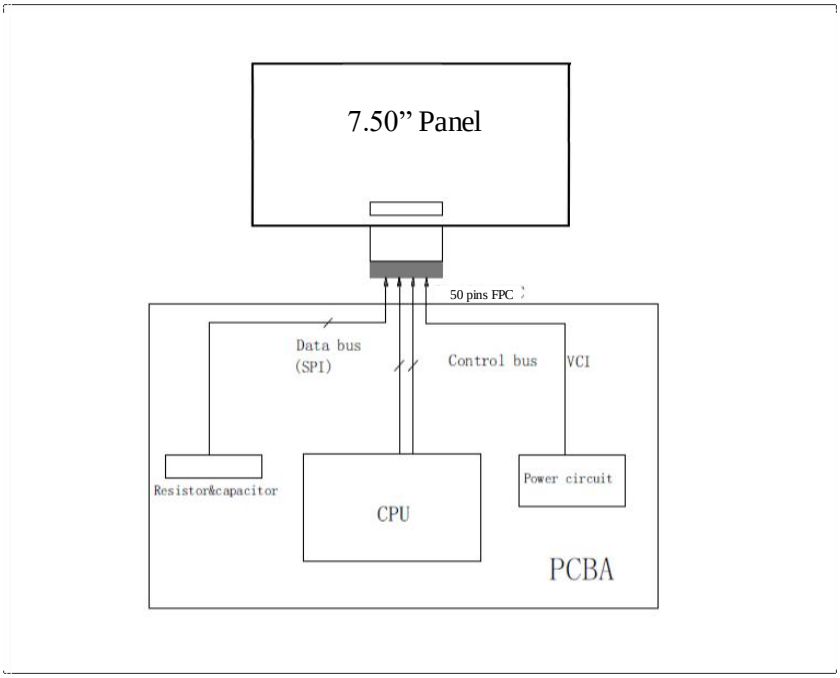
Note 14-1：Stay white pattern for storage and non-operation test.

Note 14-2：Operation is black/white/red pattern , hold time is 150S.

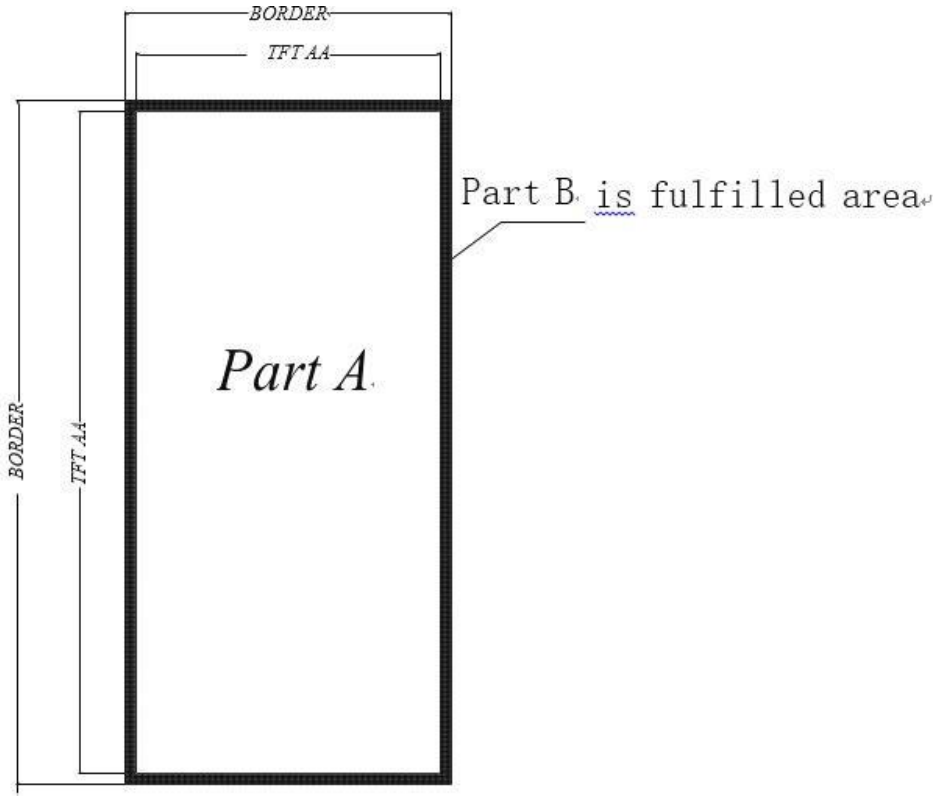
Note 14-3：The function ,appearance, opticals should meet the requirements of the test before and after the test.

Note 14-4: Keep testing after 2 hours placing at 20℃-25℃.

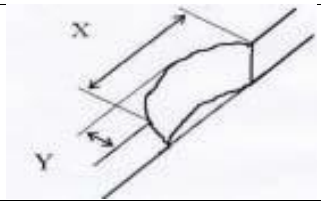
15. Block Diagram



16. PartA/PartB Specification



17. Point and line standard

Shipment Inspection Standard						
Equipment：Electrical test fixture, Point gauge						
Outline dimension	170.20(H)×111.20(V)×0.86(D)	Unit：㎜	Part-A	Active area	Part-B	Border area
Environment	Temperature	Humidity	Illuminance	Distance	Time	Angle
	19℃～25℃	55%±5% RH	600～1200Lux	300 ㎜	35Sec	
Defect type	Inspection method	Standard		Part-A		Part-B
Spot	Electric Display	D≤0.25 ㎜		Ignore		Ignore
		0.25㎜＜D≤0.4 ㎜		N≤4		Ignore
		0.4㎜＜D≤0.6 ㎜		N≤2		Ignore
		D>0.6mm		Not Allow		
Display unwork	Electric Display	Not Allow		Not Allow		Ignore
Display error	Electric Display	Not Allow		Not Allow		Ignore
Scratch or line defect(include dirt)	Visual/Film card	W≤0.2 ㎜		Ignore		Ignore
		L≤4.0mm 0.2＜W≤0.4mm		N≤2		Ignore
		L＞4 ㎜,W＞0.4㎜		Not Allow		Ignore
PS Bubble	Visual/Film card	D≤0.4mm		Ignore		Ignore
		0.4mm≤D≤0.6mm		N≤4		Ignore
		D＞0.6 mm		Not Allow		Ignore
Side Fragment	Visual/Film card	X≤3mm,Y≤0.5mm, Do not affect the electrode circuit (Edge chipping)， N≤2 X≤2mm,Y≤2mm, Do not affect the electrode circuit((Corner chipping) ,N≤2				
						

Remark	
	1.Cannot be defect & failure cause by appearance defect；
	2.Cannot be larger size cause by appearance defect；
	L=long W=wide D=point size N=Defects NO

Note 17-1：OQC inspection: One-time sampling plan for GB/T 2828.1-2012 , Inspection Level II, CR: AC/Re=0/1, MA=0.4,MI=1.0.

Note 17-2：Spot define: That only can be seen under White State or Dark State defects

Note 17-3：Any defect which is visible under gray pattern or transition process but invisible under black and white is disregarded.

Note 17-4：Any defect must be judged by Optical Microscope.

Note 17-5：Here is definition of the “Spot” and “Scratch or line defect”

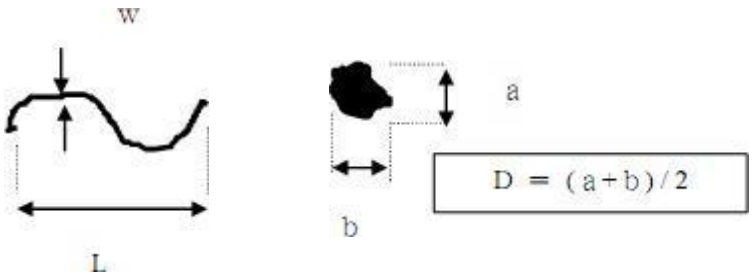
Spot: W>1/4L

Scratch or line defect :W ≤ 1/4L

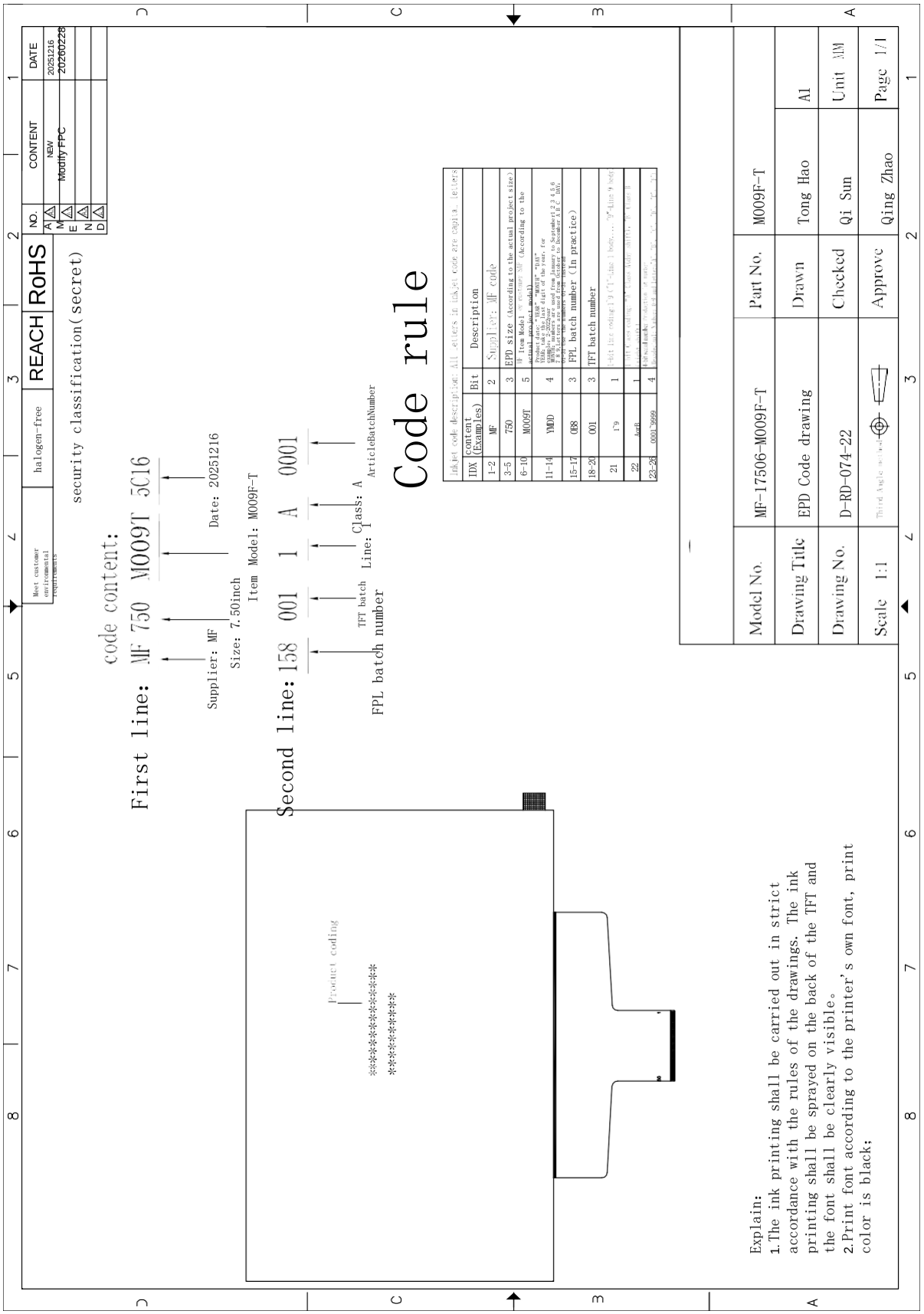
Note 17-6：Definition for L/W and D （major axis）

Note 17-7：FPC bonding area pad doesn’t allowed visual inspection

Note 17-8：



18.Barcode


Code rule
code content:
First line: MF 750 M009T 3C16
Second line: 158 001 1 A 0001
Supplier: MF
Size: 7.50inch
Item Model: M009F-T
Date: 20251216
TFT batch
FPL batch number
Line: 1
Class: A
ArticleBatchNumber
Product coding
****
****
Explain:
1.The ink printing shall be carried out in strict accordance with the rules of the drawings. The ink printing shall be sprayed on the back of the TFT and the font shall be clearly visible。
2.Print font according to the printer's own font, print color is black;
Model No.
MF-17506-M009F-T
Part No.
M009F-T
Drawing Title
EPD Code drawing
Drawn
Tong Hao
A1
Drawing No.
D-RD-074-22
Checked
Qi Sun
Unit MM
Scale 1:1
Approve
Qing Zhao
Page 1/1
NEW
Modify FPC
20251216
20260228

19.Packing

RoHS HF

	Material	Quantiy
1	Outer box	1
2	Plastic tray	11*2=22
3	Antistatic bag	2
4	Foam separator	3
5	desiccant	2*2=4
6	EPD	40*2=80
7	pallet	1/40

